0620 • PAPER 4

TOPIC

12

Experimental techniques and chemical analysis

Questions only • Original examination pages retained

1
QUESTIONS

12
TOTAL MARKS

1
SECTIONS

USE THE TRACKER • ATTEMPT WITHOUT NOTES • MARK • REVIEW ERRORS

TOPIC CONTENTS

Quick navigation and progress overview

12.1 Separation methods, titration and chemical tests

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	2020 Feb/March Paper 42 Question 4	12	☐	☐	☐

4 Iron is a typical transition element.

Iron:

- acts as a catalyst
- forms coloured compounds
- has more than one oxidation state.

(a) Name **one** major industrial process that uses iron as a catalyst and name the product made in this process.

process ....

product made ....
[2]

(b) When aqueous sodium hydroxide is added to aqueous iron(II) sulfate, a precipitate forms.

(i) What colour is this precipitate?

.... [1]

(ii) Write the ionic equation for this reaction. Include state symbols.

.... [3]

(c) Iron(II) sulfate can be converted to iron(III) sulfate by potassium manganate(VII) at room temperature.

(i) What is the role of potassium manganate(VII) in this reaction?

.... [1]

(ii) What condition must be used for this reaction to occur?

.... [1]

(iii) In terms of electron transfer, what happens to the iron(II) ions in this reaction?

.... [1]

(iv) State the colour change seen during this reaction.

from purple to .... [1]

(d) Deduce the charge on the iron ion in each of these compounds.

FeF_3 ....

$Fe(NO_3)_3$ ....
[2]

[Total: 12]